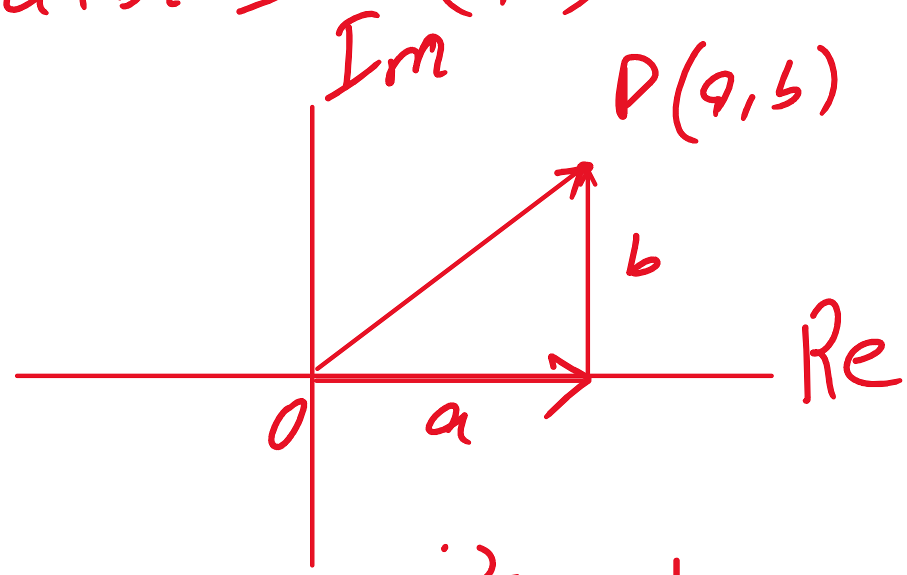
Math Class 10 Federal Board Chapter 01 Complex Numbers Exercise 1.3

$Z = a + bi = (a, b)$



$\sqrt{-1} = i \Rightarrow i^2 = -1$

$i^2 = -1$

Example: Solve $x^2 + 5 = 0$

$x^2 = -5$

$\Rightarrow x^2 = 5i^2$

$\Rightarrow \sqrt{x^2} = \pm\sqrt{5i^2}$

$\boxed{x = \pm\sqrt{5}\, i}$

$S.S = \{+\sqrt{5}i, -\sqrt{5}i\}$

$S.S = \{ +\sqrt{5}i, -\sqrt{5}i \}$

Example: Factorize:

a. z^2+9 b. $3z^2+5$

a) Z^2+9

$= Z^2-(-9)$

$= Z^2-(9i^2)$ $(\because i^2=-1)$

$= (Z)^2-(3i)^2$

Here

$Z^2+9 = \boxed{(Z-3i)(Z+3i)}$ Ans

b. $3z^2+5$

$(5x-1)$

$= 3Z^2-(-5)$

$(a)^2-(b)^2$

$= 3Z^2-5i^2$

$= 3z^2 - 5i^2$ $\quad (\sqrt{3})^2$

$= (\sqrt{3}z)^2 - (\sqrt{5}i)^2$

$= (\sqrt{3}z - \sqrt{5}i)(\sqrt{3}z + \sqrt{5}i)$ Ans

For Any Quadratic Eqn.
if roots are complex then
they must be conjugate of each other

MCQ A quadratic Eqn. has
one root of $2-3i$, then
other root will be ⟶ ?

a) $2-3i$ b) $2+3i$ c) $-2+3i$ d) $-2-3i$

Example: Mr. Waqas is an electrical engineer designing the electrical circuits for a new office building. There are three basic things to be considered in an electrical circuit. The flow of the electric current I, the resistance to that flow Z, called impedance, and electromotive force E, called voltage. Their quantities are related in the formula $E = IZ$. The current of the circuit Waqas is designing is to be $(35-40J)$ amp. Electrical engineers use the letter J to represent the imaginary unit. Find the impedance of the circuit if the voltage is to be $(430-330J)$ volts.

Given

$$E = IZ$$

$(V = IR)$

$$I = (35 - 40J) \text{ Amp.}$$

$(I = \text{Real part} + \text{Img. part } J)$

$a + bi$

$a + bj$

$$\boxed{\begin{array}{l} J = \sqrt{-1} \\ J^2 = -1 \end{array}}$$

$$E = V = (430 - 330J) \text{ volts}$$

To Find

$$Z = ??$$

Sol

As we know that

$$E = IZ$$

$$Z = \frac{E}{I} = \frac{430 - 330J}{35 - 40J}$$

$$Z = \frac{430 - 330j}{35 - 40j} \times \frac{35 + 40j}{35 + 40j}$$

$$Z = \frac{430(35 + 40j) - 330j(35 + 40j)}{(35)^2 - (40j)^2}$$

$$Z = \frac{15050 + 17200j + 13200 - 11550j}{1225 + 1600}$$

$$Z = \frac{28250 + 5650j}{2825}$$

$$Z = \frac{28250}{2825} + \frac{5650j}{2825}$$

$$\boxed{Z = 10 + 2j}$$

due to

$Z = 10 + i2$ due to capacitor Inductor

$Re(Z) = 10$ due to Resistance (R)

Example: If $z_1 = 5.5 + 5i, z_2 = 7.7 + 7i$, then calculate:

a. $z_1 + z_2$

b. $z_1 z_2$

c. $\frac{z_1}{z_2}$

a) $Z_1 + Z_2 = 5.5 + 7.7 + 5i + 7i$

$= 13.2 + 12i$

$Z_1 Z_2 = (5.5 + 5i)(7.7 + 7i)$

$= 5.5(7.7 + 7i) + 5i(7.7 + 7i)$

$= 42.35 + 38.5i - 35 + 38.5i$

$\boxed{Z_1 Z_2 = 7.35 + 77i}$ A

c) $\frac{Z_1}{}$ $\frac{5.5 + 5i}{}$

c) $\frac{z_1}{z_2} = \frac{5.5+5i}{7.7+7i}$

$= \frac{5.5+5i}{7.7+2i} \times \frac{7.7-7i}{7.7-7i}$

$= \frac{5.5(7.7-7i)+5i(7.7-7i)}{(7.7)^2+(7)^2}$

$= \frac{42.35-3\cancel{8.5}i+35+3\cancel{8}.5i}{59.29+49}$

$= \frac{77.35}{108.29}$

$= \frac{5}{7}+0i$

Example: Solve: Method #1 Equating coefficients

Example: Solve: Method #1

$$z+w=3i \rightarrow (1)$$
$$2z+3w=2 \rightarrow (2)$$

Sol:

Eq. (1) × −2 : $-2z-2w=-6i \rightarrow (3)$

eq. (2) : $2z+3w=2 \rightarrow (2)$

Adding (2) & (3), we get

$$\boxed{w=2-6i}$$

Put value of w in Eqn. (1)

$$z+2-6i=3i$$

$$z=-2+6i+3i$$

$$\boxed{z=-2+9i}$$

Thus, $\boxed{w=2-6i,\ z=-2+9i}$

Thus, $\boxed{w = 2-6i, \; z = -2+9i}$

Example: Solve:

$$z+w=3i \rightarrow (1)$$
$$2z+3w=2 \rightarrow (2)$$

$(1) \Rightarrow \boxed{z = 3i - w} \rightarrow (3)$

put (3) in (2), we get

$2(3i-w)+3w = 2$

$6i-2w+3w=2$

$\boxed{w = 2-6i}$

put value of w in Eq (3) we get

$(3) \Rightarrow z = 3i-(2-6i)$

$z = 3i-2+6i$

$\Rightarrow$ $Z = 3i - 2 + 6i$

$$\boxed{Z = -2 + 9i}$$

Methode #3

By Cramman's Rule

Example: Solve:

$$z + w = 3i$$
$$2z + 3w = 2$$

Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

$b = \begin{bmatrix} 3i \\ 2 \end{bmatrix}$ $X = \begin{bmatrix} z \\ w \end{bmatrix}$

$A_z = \begin{bmatrix} 3i & 1 \\ 2 & 3 \end{bmatrix}$, $A_w = \begin{bmatrix} 1 & 3i \\ 2 & 2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 3 - 2 = 1$$

$$\Rightarrow \boxed{|A| = 1}$$

$$|A_z| = \begin{vmatrix} 3i & 1 \\ 2 & 3 \end{vmatrix} = 9i - 2$$

$$|A_w| = \begin{vmatrix} 1 & 3i \\ 2 & 2 \end{vmatrix} = 2 - 6i$$

$$z = \frac{|A_z|}{|A|} = \frac{9i - 2}{1}$$

$$\boxed{z = -2 + 9i}$$

$$w = \frac{|A_w|}{|A|} = \frac{2 - 6i}{1} = 2 - 6i$$

$$\boxed{w = 2 - 6i}$$

Hence $\boxed{w = 2 - 6i}$ A

$\boxed{z = -2 + 5i}$

$-7 = 7 \times -1$

Solve:

1. $x^2 + 7 = 0$

$x^2 = -7$

$x^2 = 7i^2$

$\sqrt{x^2} = \pm\sqrt{7i^2}$

$\boxed{x = \pm\sqrt{7}i}$ A

$S.S = \{+\sqrt{7}i, -\sqrt{7}i\}$ A

Determine whether the given complex number is a solution of the equation.

4. $1+2i,\ x^2-2x+5=0$

5. $1-2i,\ x^2-2x+5=0$

6. $1-i,\ x^2+2x+2=0$

7. $i,\ x^2+1=0$

4. $x^2-2x+5=0 \rightarrow (1)$

$x = 1+2i$, put value of x in (1)

L.H.S. $= x^2-2x+5$

$(1+2i)^2-2(1+2i)+5$

$= 1+4i^2+2(2i)-2-4i+5$

$= 1-4+\cancel{4i}-2-\cancel{4i}+5$

$= 1-4-2+5$

$= 6-6 = 0 =$ R.H.S

7. $i, x^2 + 1 = 0$

$x = i \Rightarrow LHS = x^2 + 1$

$= i^2 + 1$

$= -1 + 1 = 0 = RHS$

Factorize the expressions:

8. $x^2 + 16$ 9. $a^2 + b^2$ 10. $x^2 + 25y^2$

(8)

$= x^2 - (-16)$

$= x^2 - (16 i^2)$

$= (x)^2 - (4i)^2$

$= \underline{(x - 4i)(x + 4i)}$ Ans

Thus factor of $x^2 + 16$ are $(x - 4i)$ & $(x + 4i)$

$(x-yi)\ \&\ (x+yi)$

9. a^2+b^2 10. x^2+25y^2

9: a^2+b^2

$= a^2-(-b^2)$

$= a^2-(b^2 i^2)$

$= a^2-(bi)^2$

$= \underline{(a-bi)(a+bi)}$

10. x^2+25y^2

$= x^2-(-25y^2)$

$= x^2-(25y^2 i^2)$

$= (x)^2 - (5yi)^2$

$= (x - 5yi)(x + 5yi)$

Solve the following system of linear equations.

11. $z - 4w = 3i$

$2z + 3w = 11 - 5i$

12. $3z + (2+i)w = 11 - i$

$(2-i)z - w = -1 + i$

12. $3z + (2+i)w = 11 - i \rightarrow (1)$

$(2-i)z - w = -1 + i \rightarrow (2)$

By Elimination method.

$\rightarrow (3)$

Eqn (2) $\times (2+i) \Rightarrow (2+i)(2-i)z - \cancel{(2+i)}w = (2+i)(-1+i)$

+

$\rightarrow (1)$

Eqn (1) $\Rightarrow$ $3z + \cancel{(2+i)}w = 11 - i$

$8z = (2+i)(-1+i) + 11 - i$

Adding Eqn (1) & (3), we get

$8z = -2 + 2i - i + i^2 + 11 - i$

$$8z = -2+2i-2+2+11\cdot$$

$$8z = -2+\cancel{2i}-\cancel{2}-1+11-\cancel{i}$$

$$8z = 8 \Rightarrow \boxed{z=1}$$

put value of z in eqn. (1, we get

$$(1) \Rightarrow 3(1)+(2+i)w = 11-i$$

$$(2+i)w = 11-i-3$$

$$w = \frac{8-i}{2+i}$$

$$w = \frac{8-i}{2+i} \times \frac{2-i}{2-i}$$

$$w = \frac{16-8i-2i+i^2}{4+1}$$

$$w = \frac{15 - 10i}{5}$$

Hence

$$\boxed{\boxed{w = 3 - 2i}}$$

$$\boxed{z = 1}$$

Alternate for finding w

put value of z in (2)

(2) $\Rightarrow$ $2 - i - w = -1 + i$

$$-w = -2 + i - 1 + i$$

$$-w = -3 + 2i$$

$$\boxed{\boxed{w = 3 - 2i}}$$

12. $3z+(2+i)w=11-i \rightarrow (1)$

$(2-i)z-w=-1+i \rightarrow (2)$

Using Substitution Method

from Eqn. (2): $-w=-(2-i)z-1+i$

$$\Rightarrow \boxed{w=(2-i)z+1-i} \rightarrow (3)$$

Put w value in Eqn (1) —

$$3z+(2+i)\left[(2-i)z+(1-i)\right]=11-i$$

Do Yourself

$$3z+5z+2+i-2i-i^2=11-i$$

$$8z+2+i-2i+1=11-i$$

$8z + 3 - \cancel{i} = 11 - \cancel{i}$

$8z = 11 - \cancel{i} - 3 + \cancel{i}$

$8z = 8 \Rightarrow \boxed{z = 1}$

Put value of z in Eq ③

$w = 2 - i + 1 - i$

$\boxed{w = 3 - 2i}$

13. In an electrical circuit, the flow of the electric current I, the impedance Z and the voltage E, are related by the formula $E = IZ$.

a. Find I, given the values:

i. $E = (70 + 220J)$ volts, $Z = (16 + 8J)$ ohms

ii. $E = (85 + 110J)$ volts, $Z = (3 - 4J)$ ohms

b. Find Z given the value:

i. $E = (-50 + 100J)$ volts, $I = (-6 - 2J)$ amp

ii. $E = (100 + 10J)$ volts, $I = (-8 + 3J)$ amp

c. Evaluate $\frac{1}{z - z^2}$ when $z = \frac{1-i}{10}$

$J = i$

$J^2 = -1$

$Z = \frac{E}{I}$

a) ii) $I = ?$

As $E = IZ$

$$I = \frac{E}{Z} = \frac{85+110J}{3-4J}$$

$$I = \frac{85+110J}{3-4J} \times \frac{3+4J}{3+4J}$$

$$I = \frac{-185+670J}{3^2+4^2}$$

$$I = \frac{-185+670J}{25}$$

$$\boxed{I = \left(\frac{-37}{5} + \frac{134}{5}J\right) \text{Amps}}$$

A

b. Find Z given the value:

i. $E=(-50+100J)$ volts, $I=(-6-2J)$ amp

ii. $E=(100+10J)$ volts, $I=(-8+3J)$ amp

i) $E = IZ$

$$Z = \frac{E}{I} = \frac{-50+100J}{-6-2J}$$

$$Z = \frac{-50+100J}{-6-2J} \times \frac{-6+2J}{-6+2J}$$

$$\boxed{Z = \left(\frac{5}{2} - \frac{35}{2}J\right) \text{ ohms}}$$

Impedance

c. Evaluate $\dfrac{1}{z-z^2}$ when $z=\dfrac{1-i}{10}$

$$z^2 = \left(\frac{1-i}{10}\right)^2 = \frac{(1-i)^2}{10^2}$$

$$Z^2 = \frac{1 + i^2 - 2i}{100}$$

$$Z^2 = \frac{\cancel{1} - \cancel{1} - 2i}{100}$$

$$\boxed{Z^2 = \frac{-i}{50}}$$

Now $$\frac{1}{Z - Z^2} = \frac{1}{\frac{1-i}{10} - \left(\frac{-i}{50}\right)}$$

$$= \frac{1}{\frac{1-i}{10}(5) + \frac{i}{50}}$$

$$= \frac{1}{5(1-i) + i}$$

$$50$$

$$= \frac{1}{\frac{5-5i+i}{50}} = \frac{1}{\frac{5-4i}{50}}$$

$$= 1 \div \frac{5-4i}{50}$$

$$= 1 \times \frac{50}{5-4i}$$

$$\frac{1}{z-z^2} = \frac{50}{5-4i} \times \frac{5+4i}{5+4i}$$

$$= \frac{250+200i}{5^2+4^2}$$

$$\frac{1}{z-z^2} = \frac{250+200i}{41}$$

$$\boxed{\frac{1}{z-z^2} = \frac{250}{41} + \frac{200}{41}i}$$

End of Exercise 1.3